

The asymmetric effects of exchange rate (USD/VND) on Vietnam's trade balance: Evidence from Nonlinear ARDL and Machine Learning models (2015-2025)

Bui Viet Huy

DSEB-MFE, NEU

Email: 11230542@st.neu.edu.vn

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Lecturer: Tran Thi Ha

Abstract. This study investigates the asymmetric impact of the USD/VND real exchange rate (RER) on Vietnam's trade dynamics from January 2015 to December 2025. Utilizing a Nonlinear ARDL (NARDL) framework, we decompose RER movements to capture asymmetric transmission across three parallel models: aggregate trade balance, exports, and imports. The research advances existing literature by incorporating the post-COVID-19 recovery phase, oil price controls, and benchmarking NARDL's predictive performance against machine learning architectures (LSTM and Random Forest). Empirical findings confirm the J-curve effect and significant asymmetry, with VND depreciation exerting a more pronounced long-term impact on the trade balance than appreciation. The COVID-19 pandemic is identified as a significant structural disruptor, while robustness checks using REER validate the core results. Policy implications emphasize a 6–12 month adjustment lag for central bank interventions and the necessity for asymmetric risk-hedging strategies among export enterprises to mitigate non-linear currency shocks.

Key words: Exchange rate; Trade balance; NARDL; J-curve effect; Asymmetric effects; Vietnam; COVID-19; Machine learning

I. Introduction

Vietnam's integration into the global economy represents a remarkable trajectory of export-led growth, with total trade turnover projected to reach approximately \$405 billion by 2024, positioning the nation among the world's top 25 merchandise exporters. Central to this economic strategy is the managed floating exchange rate regime, where the USD/VND bilateral rate serves as the primary instrument of monetary transmission. The disproportionate significance of this specific currency pair stems from the fact that approximately 90% of Vietnam's international trade is invoiced and settled in US dollars. Consequently, the USD/VND real bilateral exchange rate (RER) functions as the definitive mechanism governing the international price competitiveness of Vietnamese commodities, effectively mediating the relationship between domestic policy and global trade outcomes.

However, the exchange rate–trade nexus has encountered unprecedented turbulence during the 2015–2025 period. The dual shocks of the COVID-19 pandemic and the subsequent aggressive tightening cycle by the US Federal Reserve exerted sustained depreciation pressure on the Vietnamese dong, culminating in historic lows and significant central bank interventions by early 2025. Despite the criticality of these dynamics, the existing empirical literature exhibits notable gaps. Prior research, including prominent studies by Truong and Vo (2023) [1] and Nguyen et al. (2024) [3], either fails to encompass the full post-pandemic recovery phase or omits vital control variables such as global oil prices and Foreign Direct Investment (FDI) within a unified framework. Furthermore, the absence of a simultaneous estimation across aggregate trade, export, and import dynamics prevents a holistic understanding of how asymmetric transmission mechanisms operate across different trade components.

To address these limitations, this study investigates three fundamental research questions:

- (1) Do the asymmetric effects of the real exchange rate (RER) on Vietnam's trade balance persist and intensify when accounting for recent structural shocks?
- (2) Is the J-curve hypothesis validated for Vietnam over the extended 2015–2025 sample period?
- (3) Do machine learning models, specifically Random Forest and Gradient Boost, achieve superior predictive accuracy compared to the traditional Nonlinear Autoregressive Distributed Lag (NARDL) framework?

Utilizing a rigorous dataset of 132 monthly observations, the study employs the NARDL approach to decompose exchange rate movements into positive and negative partial sums, while systematically benchmarking econometric results against machine learning models to evaluate the trade-off between structural interpretability and predictive validity.

This research provides four distinct contributions to the empirical literature on exchange rate economics. First, it extends the temporal scope to the end of 2025, offering the first comprehensive analysis of Vietnam's trade dynamics that incorporates the post-COVID-19 stabilization era. Second, by introducing a structural dummy for the pandemic and integrating WTI crude oil prices as a control variable, the model effectively captures the impact of energy import dependence and cost-push inflation on trade flows. Third, the simultaneous estimation of three parallel NARDL models aggregate trade balance, exports, and imports enables a direct comparison of exchange rate asymmetry across all trade facets. Finally, by bridging the gap between econometric modeling and modern forecasting algorithms, this study provides policymakers with evidence-based insights into the relative merits of machine learning versus traditional structural models for trade balance management in an increasingly volatile global environment.

II. Review of Relevant Literature

1. Theoretical Foundations of the Exchange Rate-Trade Balance Nexus

The classical starting point is the elasticity approach, especially the Marshall-Lerner condition. According to this condition, a currency depreciation improves the trade balance only when the combined price elasticities of export and import demand exceed one [9, 7, 8]. This means that exchange rate policy is effective only if trade volumes respond sufficiently to changes in relative prices [1, 2].

In Vietnam, this mechanism is not automatic because exports are strongly linked to foreign-invested manufacturing, where imported intermediate inputs account for a large share of production [20, 1]. As a result, depreciation may raise the domestic-currency value of exports without generating a proportional increase in export volume [2, 3].

The J-curve hypothesis provides a dynamic explanation of why depreciation may first worsen the trade balance before improving it later [21, 10, 11]. In the short run, existing contracts and slow quantity adjustment mean that import costs rise faster than export volumes respond [4, 22]. Over time, firms and consumers adjust to the new price structure, allowing exports to expand and imports to decline [23]. This framework is particularly relevant for Vietnam because the economy is open, import-dependent and highly integrated into global value chains [24, 2].

Other theoretical perspectives also support the need for a broader exchange-rate framework. The purchasing power parity tradition treats the exchange rate as a long-run price-level adjustment mechanism [6], while the Balassa-Samuelson effect explains why fast-growing developing economies may experience real appreciation as productivity rises [11, 12]. The absorption approach adds a macroeconomic perspective by arguing that depreciation improves the trade balance only if national income rises relative to domestic absorption [13, 14].

Together, these theories suggest that the exchange rate-trade balance relationship depends not only on prices, but also on production structure, import dependence, domestic demand and long-run competitiveness [15, 16].

2. Methodological Development: From ARDL to NARDL

The ARDL bounds testing approach developed by Pesaran, Shin and Smith (2001) [6] is widely used in trade-balance studies because it can be applied when variables are a mixture of $I(0)$ and $I(1)$, performs well in small samples and estimates both short-run and long-run relationships within a single framework. These features make ARDL suitable for developing economies where macroeconomic time series are often limited and mixed in their integration properties.

However, the standard linear ARDL model assumes symmetry: depreciation and appreciation are treated as equal in magnitude but opposite in direction. This assumption is restrictive because firms, consumers and policymakers may react differently to exchange rate increases and decreases. For example, exporters may not expand immediately after depreciation if they rely heavily on imported inputs, while importers may react more quickly to appreciation because foreign goods become cheaper.

The NARDL framework proposed by Shin, Yu and Greenwood-Nimmo (2014) [5] directly addresses this limitation by decomposing the exchange rate into positive and negative partial sums. This allows depreciation and appreciation to enter the model separately, so their short-run and long-run effects can differ. The framework also allows researchers to test asymmetry using Wald tests and to trace adjustment paths through dynamic multipliers. For studies of Vietnam's trade balance, NARDL is especially useful because it can test whether the J-curve is asymmetric rather than assuming a single uniform response to exchange rate changes.

3. Empirical Evidence on Exchange Rate Effects

International evidence generally supports the view that exchange rate effects are asymmetric. Bahmani-Oskooee and Fariditavana (2015)[10] show that US bilateral trade balances respond differently to depreciation and appreciation, while Bahmani-Oskooee and Aftab (2017)[16] find that asymmetric effects are clearer at the industry level than at the aggregate level. This suggests that aggregate models may hide sector-specific responses and underestimate the true importance of exchange rate movements. Evidence from ASEAN and emerging economies is also mixed but broadly consistent with the need for nonlinear modelling. Studies on Thailand, Malaysia, Indonesia and China show that the size, timing and direction of exchange-rate effects vary across countries, partners and industries. A recurring finding is that trade invoicing and vehicle-currency effects matter: when trade is invoiced mainly in USD, bilateral exchange rates alone may not fully capture the relevant price channel. This is important for Vietnam because the USD plays a dominant role in trade settlement and policy signalling.

4. Evidence from Vietnam

Vietnam provides a suitable context for studying asymmetric exchange-rate effects because its exchange rate regime is managed rather than freely floating, its export sector is strongly FDI-driven, and its trade structure is highly dependent on imported inputs. During 2015-2025, Vietnam experienced several exchange-rate and external shocks, including the 2015 RMB devaluation, US-China trade tensions, the COVID-19 period, the 2020 US currency-manipulator designation and the 2022-2024 Fed tightening cycle. These events make it unlikely that the exchange rate-trade balance relationship remained stable and symmetric throughout the whole period. Early studies on Vietnam mainly used VAR, IRF and linear ARDL models. Phan and Jeong (2015)[4], S. A. Nusair (2017) [13], B. N. Iyke and S. H. Ho (2018) [15] generally find evidence of a J-curve pattern: depreciation tends to worsen the trade balance in the short run before improvement appears later. These findings provide an important foundation, but they are limited because linear models impose symmetry between depreciation and appreciation. More recent NARDL studies provide stronger evidence of asymmetry. L. D. Truong and D. Van Vo (2023) [1] find that VND depreciation worsens the trade balance in the short run but improves it in the long run, while appreciation effects are weaker or statistically insignificant. H. H. G. Bao and H. P. Le(2021) [2]

show that including the USD as a vehicle currency increases the number of significant exchange-rate effects in Vietnam-EU trade, confirming the importance of USD-based channels. L. H. Phong, H. H. G. Bao, and D. T. B. Van (2018) [23] also find that industry-level NARDL models reveal asymmetric responses that are not visible in aggregate ARDL models. Overall, the Vietnam literature supports the use of NARDL and highlights the risks of relying only on aggregate linear specifications.

5. Machine Learning as a Complementary Forecasting Approach

Although NARDL is valuable for identifying asymmetric long-run and short-run relationships, it remains limited by its pre-specified functional form and relatively small set of regressors [5, 6]. It is less flexible in capturing complex interactions among exchange rates, industrial production, FDI, external demand and global price conditions [1, 2, 3].

Machine learning methods can complement NARDL by improving predictive performance and detecting nonlinear relationships that traditional econometric models may miss [17, 19, 18]. Linear Regression can serve as a transparent benchmark model, while Random Forest and XGBoost are more flexible ensemble methods capable of capturing nonlinear effects and interactions among variables [18, 19].

Recent macroeconomic forecasting studies show that these models often outperform traditional linear time-series models, especially under uncertainty and structural instability [17].

The main concern with machine learning is interpretability. This can be addressed through explainable AI tools such as SHAP values, which decompose each prediction into feature-level contributions [19]. In this study, SHAP can help evaluate whether exchange rate depreciation, appreciation, REER, industrial production, FDI and other macroeconomic variables are truly important predictors of Vietnam's trade balance [1, 2]. Therefore, the combined NARDL-machine learning approach allows the study to balance structural explanation with predictive flexibility [5].

6. Research Gap and Contribution

The literature reveals four main gaps. First, most NARDL studies on Vietnam use data ending before 2021, meaning they do not fully capture the COVID-19 shock, the currency-manipulator episode and the Fed tightening cycle. Extending the sample to 2015-2025 allows the present study to analyse a more recent and structurally different period. Second, previous Vietnam studies have not systematically integrated NARDL with machine learning. This study contributes by combining an econometric model that identifies asymmetric exchange-rate effects with forecasting models that capture nonlinear interactions. Third, the role of REER remains underexplored compared with the bilateral USD/VND rate, even though Vietnam trades extensively with China, Korea, the EU and other non-USD partners. Including both USD/VND and REER therefore strengthens the analysis of external competitiveness. Fourth, earlier studies rarely examine whether asymmetry changes across sub-periods. This study addresses that gap by considering structural breaks and rolling-window dynamics.

III. Data and Methodology

1. Data Sources and Variable Description

This study employs monthly time-series data for Vietnam over the period from January 2015 to December 2025, yielding a total of 132 observations. The monthly frequency is selected because the exchange rate–trade balance relationship is expected to involve short-run adjustment lags, particularly under the J-curve mechanism. Compared with quarterly or annual data, monthly observations allow the model to capture adjustment dynamics within the 3–12 month window, during which import contract rigidity, delayed export responses, and gradual quantity adjustments may occur.

Table 1: Variable Description and Source

Variable	Type	Description	Source
FDI (billion USD)	float	Foreign Direct Investment in billion USD	GSO
EX (USD)	float	Export value in USD	GSO
IM (USD)	float	Import value in USD	GSO
CPI_US (2010 = 100)	float	Consumer Price Index of the United States, base year 2010	IMF
CPI_VN (2010 = 100)	float	Consumer Price Index of Vietnam, base year 2010	IMF
REER (2007M12 = 100)	float	Real Effective Exchange Rate, base December 2007	IMF
NER	float	Nominal Exchange Rate	WB
IPI_VN (2015 = 100)	float	Industrial Production Index of Vietnam, base year 2015	IMF
IPI_World (2015 = 100)	float	Industrial Production Index of the World, base year 2015	IMF
COVID	dummy	Dummy variable for the COVID-19 pandemic period	GSO
WTI Spot Price FOB	float	West Texas Intermediate crude oil spot price per barrel	IMF
M2 (billion VND)	integer	Broad money supply in billion VND	WB

The dependent variable in the baseline model is the trade balance, measured as the ratio of exports to imports. Export value and import value are measured in USD and obtained from GSO[25]. A trade balance ratio greater than one indicates that exports exceed imports, while a value below one indicates a trade deficit. In logarithmic terms, a positive value of $\ln TB_t$ reflects a trade surplus, whereas a negative value suggests that imports exceed exports. In addition to the aggregate trade balance model, this study estimates separate export and import models to identify whether exchange rate movements affect the trade balance through the export channel, the import channel, or both.

The key explanatory variable is the exchange rate. This study uses both the nominal exchange rate and the real effective exchange rate to capture different dimensions of exchange rate movements. The nominal exchange rate reflects the VND/USD exchange rate and is obtained from the World Bank[27], while the real effective exchange rate is measured with December 2007 as the base period and obtained from the IMF[26]. The real exchange rate indicator is then decomposed into positive and negative partial-sum components in the NARDL framework to examine whether exchange rate appreciation and depreciation generate asymmetric effects on Vietnam's trade balance.

Several macroeconomic control variables are included to reduce omitted-variable bias. Foreign direct investment, measured in billion USD, captures external capital inflows and production capacity. The Consumer Price Index of Vietnam and the Consumer Price Index of the United States, both with 2010 as the base year, are used to account for relative price movements. Industrial Production Index data for Vietnam and the world, with 2015 as the base year, proxy domestic and external economic activity, respectively. WTI crude oil spot price is included to control for global energy price fluctuations, while broad money supply, measured in billion VND, captures domestic monetary conditions. Finally, a COVID-19 dummy variable is added to control for pandemic-related disruptions during the sample period.

Table 2: Descriptive Statistics for All Variables

Variable	Mean	Std.	Min	Max	Skew.	Kurt.
FDI (billion USD)	1.7090	0.4361	1.0090	2.9576	0.7157	0.1364
EX (USD)	29187349.31	9410885.56	12096714.00	51226377.00	0.4343	-0.6910
IM (USD)	26842165.55	8449335.93	12352512.00	50115711.00	0.4925	-0.6278
CPI_US (2010 = 100)	124.7342	13.6643	107.1778	148.9529	0.4734	-1.3258
CPI_VN (2010 = 100)	168.7185	16.1529	143.7558	198.6607	0.1396	-1.0878
REER (2007M12 = 100)	147.1920	5.1064	136.7000	161.8300	0.3577	-0.3641
NER	23427.9876	1169.4820	21335.2000	26386.9091	1.0079	0.4347
IPI_VN (2015 = 100)	100.7669	12.6400	69.7494	128.8706	0.0212	-0.6418
IPI World (2015 = 100)	113.2103	8.5611	95.6055	128.2156	-0.0147	-1.1474
COVID	0.3485	0.4783	0.0000	1.0000	0.6433	-1.6108
WTI Spot Price FOB (USD/Barrel)	62.3897	17.4428	16.5500	114.8400	0.3795	0.3801
M2 (billion VND)	11531702.66	4189393.87	5192194.00	20598667.00	0.3143	-0.8821

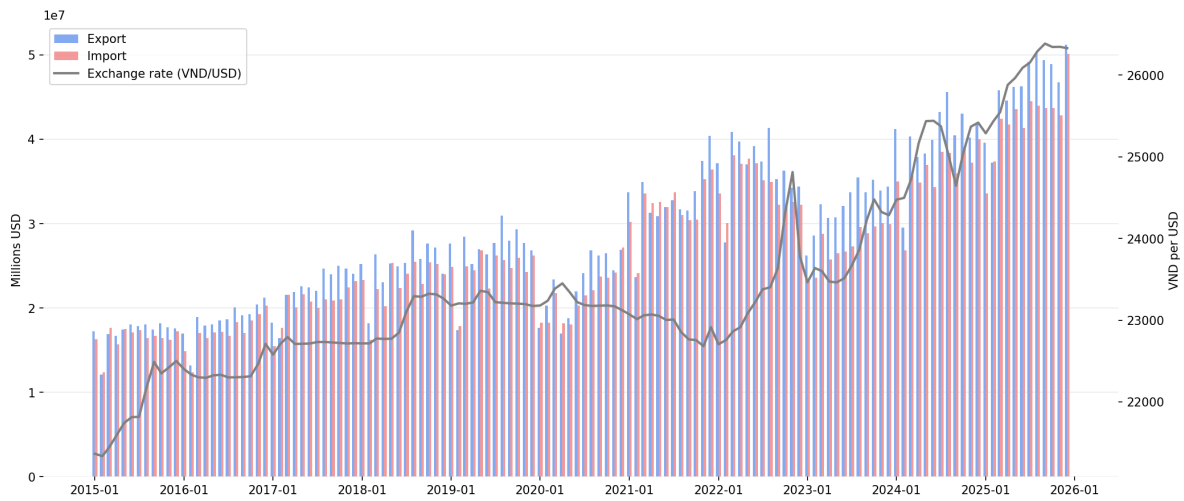


Figure 1. Vietnam's exports, imports, and nominal exchange rate from January 2015 to December 2025.

2. Descriptive Statistics

Table 2 reports the descriptive statistics of all variables, including mean, standard deviation, minimum, maximum, skewness, and kurtosis. The results provide an initial overview of the distributional characteristics of the dataset before formal econometric estimation.

The trade variables show substantial variation over the sample period. Export value records a mean of 29.19 million USD, ranging from 12.10 million USD to 51.23 million USD, while import value has a slightly lower mean of 26.84 million USD, with a range from 12.35 million USD to 50.12 million USD. This suggests that both export and import activities fluctuated considerably during the period under study.

Figure 1 further illustrates the movement of exports, imports, and the nominal exchange rate over time. The figure visually confirms the strong upward trend in Vietnam's trade flows, especially in the later years of the sample period. Exports and imports generally move closely together, indicating that Vietnam's trade balance is shaped by simultaneous changes in both trade channels rather than by only one side of trade activity. The nominal exchange rate also shows an overall increasing pattern, suggesting a gradual depreciation of the Vietnamese dong against the US dollar during the sample period.

The exchange rate indicators also display noticeable movements. The nominal exchange rate has a

mean of 23,427.99 and ranges from 21,335.20 to 26,386.91, with positive skewness of 1.0079, indicating a relatively longer right tail. Meanwhile, the REER has a mean of 147.1920 and a lower standard deviation of 5.1064, suggesting more moderate dispersion compared with trade and monetary variables.

Regarding macroeconomic and control variables, FDI averages 1.7090 billion USD, while WTI oil prices show considerable volatility, ranging from 16.55 to 114.84 USD per barrel. The M2 variable also exhibits a wide range, increasing from 5,192,194 to 20,598,667 billion VND, which reflects the expansion of money supply over the sample period. The COVID dummy has a mean of 0.3485, indicating that around one-third of the observations fall within the COVID-affected period.

In terms of distributional shape, most variables show moderate skewness. However, NER is more strongly right-skewed, while IPI World is nearly symmetric, with skewness close to zero. The kurtosis values are generally negative for several variables, such as CPI_US, CPI_VN, IPI World, and COVID, suggesting relatively flatter distributions. Overall, the descriptive statistics indicate that the dataset contains meaningful variation across trade, exchange rate, price, production, and macroeconomic indicators.

3. Real Bilateral Exchange Rate (RER) Definition and Computation

The key explanatory variable in this study is the real bilateral exchange rate between the Vietnamese dong and the US dollar [1, 2, 3]. It is calculated by adjusting the nominal exchange rate for price level differences between the two countries, capturing the purchasing power of the currency. Specifically, the real exchange rate at time t is defined as:

$$RER_t = NER_t \left(\frac{CPI_US_t}{CPI_VN_t} \right), \quad (1)$$

where NER_t denotes the monthly average nominal VND/USD exchange rate, CPI_US_t is the US consumer price index, and CPI_VN_t is Vietnam's consumer price index. Both CPI series are rebased to 2015 = 100 to standardize the price levels before computing the real exchange rate [6, 5].

An increase in RER_t indicates a real depreciation of the Vietnamese dong relative to the US dollar, meaning that Vietnamese goods become relatively cheaper for foreign buyers. Conversely, a decrease in RER_t signals a real appreciation of the dong, making Vietnamese goods more expensive in international markets. This measure serves as the central explanatory variable in the NARDL and ARDL models used to analyze the impact of exchange rate movements on trade balance, exports, and imports in Vietnam from 2015 to 2025 [1, 2, 3].

4. Theoretical Model: Two-Country Trade Framework

Following the standard two-country trade model, the trade balance is theoretically determined by the real exchange rate, domestic economic activity, and foreign demand [1, 2, 3, 4]. In this framework, domestic and foreign goods are imperfect substitutes, and trade flows respond to relative prices and income levels. The reduced-form trade balance equation can be written as:

$$\ln TB_t = f(\ln RER_t, \ln Y_{VN,t}, \ln Y_{W,t}). \quad (2)$$

where TB_t is the trade balance ratio, RER_t is the real exchange rate, $Y_{VN,t}$ represents domestic income, and $Y_{W,t}$ represents foreign income or world demand [6, 5]. In this framework, the real exchange rate captures the relative price effect, domestic income captures import demand pressure, and foreign income captures external demand for Vietnamese exports [1, 2].

A rise in the real exchange rate represents real depreciation of the Vietnamese dong. Theoretically, depreciation may improve the trade balance by making exports cheaper and imports more expensive. However, this effect may not be immediate because exports and imports may respond differently to

favourable and unfavourable exchange rate movements [7, 9]. Therefore, imposing symmetry may produce biased or incomplete estimates of the exchange rate–trade balance relationship [5].

5. Nonlinear ARDL Model and Partial Sum Decomposition

To address the limitation of symmetry in the linear ARDL framework, this study applies the nonlinear autoregressive distributed lag model developed by Shin et al. (2014)[5]. The NARDL model allows the real exchange rate to be decomposed into positive and negative partial sums, thereby capturing asymmetric long-run and short-run effects. The positive partial sum represents real depreciation of the Vietnamese dong, while the negative partial sum represents real appreciation.

Following Shin et al. (2014)[5], changes in the real exchange rate are decomposed as follows:

$$\ln_RER_pos_t = \sum_{j=1}^t \max(\Delta \ln RER_j, 0). \quad (3)$$

$$\ln_RER_neg_t = \sum_{j=1}^t \min(\Delta \ln RER_j, 0). \quad (4)$$

This decomposition allows real depreciation and real appreciation to have different long-run and short-run effects on the trade balance, exports, and imports.

6. Model Specification

To keep the empirical equations compact within the two-column format, the control variables are grouped into the following vector:

$$Z_t = [\ln FDI_t, IPI_VN_t, IPI_World_GR_diff_t, \ln_M2_diff_t, \ln_WTI_t, COVID_t]. \quad (5)$$

The first empirical model uses the trade balance as the dependent variable:

$$\ln TB_t = \alpha_0 + \alpha_1 \ln_RER_pos_t + \alpha_2 \ln_RER_neg_t + \delta' Z_t + \varepsilon_t. \quad (6)$$

The second empirical model uses exports as the dependent variable:

$$\ln EX_t = \beta_0 + \beta_1 \ln_RER_pos_t + \beta_2 \ln_RER_neg_t + \phi' Z_t + u_t. \quad (7)$$

The third empirical model uses imports as the dependent variable:

$$\ln IM_t = \gamma_0 + \gamma_1 \ln_RER_pos_t + \gamma_2 \ln_RER_neg_t + \psi' Z_t + v_t. \quad (8)$$

In these models, $\ln_RER_pos_t$ measures the effect of real depreciation, while $\ln_RER_neg_t$ measures the effect of real appreciation. The use of separate positive and negative exchange rate components allows the study to examine whether depreciation and appreciation have asymmetric effects on Vietnam's trade balance, exports, and imports.

7. Expanded Model Specification

For clarity, the three models can also be written in expanded form. The trade balance model is specified as follows:

$$\ln TB_t = \alpha_0 + \alpha_1 \ln_RER_pos_t + \alpha_2 \ln_RER_neg_t + \alpha_3 \ln FDI_t + \alpha_4 IPI_VN_t + \alpha_5 IPI_World_GR_diff_t + \alpha_6 \ln_M2_diff_t + \alpha_7 \ln_WTI_t + \alpha_8 COVID_t + \varepsilon_t. \quad (9)$$

The export model is specified as follows:

$$\ln EX_t = \beta_0 + \beta_1 \ln_RER_pos_t + \beta_2 \ln_RER_neg_t + \beta_3 \ln FDI_t + \beta_4 IPI_VN_t + \beta_5 IPI_World_GR_diff_t + \beta_6 \ln_M2_diff_t + \beta_7 \ln_WTI_t + \beta_8 COVID_t + u_t. \quad (10)$$

The import model is specified as follows:

$$\ln IM_t = \gamma_0 + \gamma_1 \ln_RER_pos_t + \gamma_2 \ln_RER_neg_t + \gamma_3 \ln FDI_t + \gamma_4 IPI_VN_t + \gamma_5 IPI_World_GR_diff_t + \gamma_6 \ln_M2_diff_t + \gamma_7 \ln_WTI_t + \gamma_8 COVID_t + v_t. \quad (11)$$

8. General NARDL Error Correction Specification

For each dependent variable $Y_t \in \{\ln TB_t, \ln EX_t, \ln IM_t\}$, the general NARDL error correction specification is written as:

$$\Delta Y_t = c + \rho Y_{t-1} + \theta_1 \ln_RER_pos_{t-1} + \theta_2 \ln_RER_neg_{t-1} + \theta'_3 Z_{t-1} + \sum_{i=1}^p \lambda_i \Delta Y_{t-i} + \sum_{i=0}^{q_1} \pi_i^{pos} \Delta \ln_RER_pos_{t-i} + \sum_{i=0}^{q_2} \pi_i^{neg} \Delta \ln_RER_neg_{t-i} + \sum_{i=0}^{q_3} \omega'_i \Delta Z_{t-i} + e_t. \quad (12)$$

where Y_t represents the dependent variable in each of the three models, and Z_t denotes the vector of control variables. The parameters θ_1 and θ_2 capture the long-run asymmetric effects of real depreciation and real appreciation, while π_i^{pos} and π_i^{neg} capture their short-run asymmetric effects.

9. Estimation Procedure

The estimation procedure follows an extended Box-Jenkins logic, beginning with the examination of time-series properties, followed by lag selection, cointegration testing, nonlinear ARDL estimation, and diagnostic checking. This sequential procedure ensures that the empirical model is statistically valid and that the estimated coefficients can be interpreted reliably.

IV. Results

1. Unit Root Tests

Table 3: Unit Root Test Results

Variable	ADF Statistic	PP Statistic	Order
ln_FDI	-0.598 (0.871)	-3.229** (0.018)	Mixed I(0)/I(1)
ln_TB	-5.103*** (0.000)	-9.342*** (0.000)	I(0)
ln_EX	-1.029 (0.743)	-2.133 (0.231)	I(1)
ln_IM	-1.268 (0.644)	-1.648 (0.458)	I(1)
ln_RER_pos	-7.235*** (0.000)	-6.739*** (0.000)	I(0)
ln_RER_neg	-4.890*** (0.000)	-8.397*** (0.000)	I(0)
IPI_VN	-1.654 (0.455)	-2.775* (0.062)	I(1)
IPI_World_diff	-6.797*** (0.000)	-36.224*** (0.000)	I(0)
ln_WTI	-2.320 (0.166)	-2.340 (0.159)	I(1)
ln_M2_diff	-2.027 (0.275)	-11.744*** (0.000)	Mixed I(0)/I(1)
COVID	-1.483 (0.542)	-1.608 (0.480)	I(1)

Note: Values in parentheses are p-values. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

The first step is to conduct unit root tests for all variables. This study applies both the Augmented Dickey-Fuller test and the Phillips-Perron test to examine whether each variable is stationary at level, $I(0)$, or becomes stationary after first differencing, $I(1)$. This step is essential because the ARDL and NARDL frameworks are valid only when the variables are integrated of order zero or one. If any variable is integrated of order two, $I(2)$, the bounds testing procedure is no longer appropriate.

The null hypothesis of both the ADF and PP tests is that the series contains a unit root. Rejection of the null hypothesis indicates stationarity. As reported in Table 3, the results confirm that none of the variables is integrated of order two, $I(2)$, which satisfies the key precondition for applying the ARDL and NARDL bounds testing procedures.

Several variables are stationary at level. In particular, $\ln TB$, $\ln RER_{pos}$, $\ln RER_{neg}$, and IPI_{World_diff} reject the unit root null hypothesis under both the ADF and PP tests. This indicates that these variables are integrated of order zero, $I(0)$. The stationarity of the positive and negative partial-sum exchange rate components is especially relevant for the NARDL specification, as it allows the model to examine asymmetric exchange rate effects without violating the integration-order requirement.

By contrast, $\ln EX$, $\ln IM$, $\ln WTI$, and $COVID$ do not reject the unit root null hypothesis at level and are therefore treated as $I(1)$. The Vietnamese industrial production index, IPI_{VN} , is also classified as $I(1)$, although the PP statistic is weakly significant at the 10% level. This suggests that the evidence of level stationarity is not strong enough across both tests, so the variable is conservatively treated as first-difference stationary.

Two variables show mixed evidence across the ADF and PP tests. For $\ln FDI$, the ADF test fails to reject the unit root null, while the PP test rejects it at the 5% level. Similarly, $\ln M2_diff$ is non-stationary according to the ADF test but stationary according to the PP test at the 1% level. These results are reported as mixed $I(0)/I(1)$ cases. Such mixed integration orders are acceptable within the ARDL and NARDL frameworks, provided that no variable is $I(2)$.

Overall, the unit root results support the use of the ARDL and NARDL approaches. The variables consist of a combination of $I(0)$, $I(1)$, and mixed $I(0)/I(1)$ series, while no evidence of second-order integration is found. Therefore, the analysis can proceed to lag selection and bounds testing for the long-run relationship among the variables.

2. Optimal Lag Selection

The next step is to determine the optimal lag structure for each ARDL specification. Appropriate lag selection is important because the ARDL and NARDL frameworks are designed to capture both short-run adjustment dynamics and long-run equilibrium relationships among variables. Selecting too few lags may lead to omitted dynamic effects and autocorrelated residuals, while selecting too many lags may reduce estimation efficiency and consume degrees of freedom. In this study, the optimal lag orders are selected based on the Akaike Information Criterion (AIC), which is widely used in time-series analysis due to its ability to balance model fit and parsimony.

Table 4: Optimal Lag Selection for ARDL Models

Model	Dep. Var.	Optimal ARDL Order	AIC	BIC	HQIC
Model 1	$\ln TB$	ARDL(5, 0, 6, 0, 6)	-371.887	-306.653	-345.384
Model 2	$\ln EX$	ARDL(6, 3, 1, 6, 6, 5, 2)	-248.573	-143.925	-206.060
Model 3	$\ln IM$	ARDL(3, 3, 1, 2, 2, 4, 3)	-284.128	-209.975	-253.999

Table 4 reports the optimal lag structures for the three ARDL specifications, namely the trade balance model, export model, and import model. The results indicate that the optimal lag orders differ across models, suggesting that each trade-related variable follows a distinct adjustment process.

For the trade balance model, the selected specification is ARDL(5, 0, 6, 0, 6). This implies that the trade balance depends strongly on its own historical values, while several explanatory variables also affect the trade balance with relatively long lag structures. Among the three models, this specification records the lowest AIC value, indicating the strongest overall in-sample fit according to the Akaike criterion.

For the export model, the optimal specification is ARDL(6, 3, 1, 6, 6, 5, 2), which represents the most lag-intensive structure among the three equations. This suggests that export responses to exchange rate movements and macroeconomic conditions are relatively gradual and persistent over time. By contrast, the import model adopts the specification ARDL(3, 3, 1, 2, 2, 4, 3), indicating a shorter but still dynamic adjustment process for imports.

Overall, the lag selection results confirm that the relationship between exchange rate movements and Vietnam's trade performance is dynamic rather than instantaneous. The selected lag structures provide the basis for the subsequent ARDL and NARDL estimations.

Table 5: ARDL Estimation Results for the Trade Balance Model

Variable	Coef.	Std. Err.	z	P> z	[0.025	0.975]
const	0.0493	0.040	1.240	0.218	-0.030	0.128
ln_TB.L1	0.2157	0.092	2.350	0.021	0.034	0.398
ln_TB.L2	0.1109	0.101	1.103	0.273	-0.089	0.310
ln_TB.L3	0.2402	0.098	2.462	0.015	0.047	0.434
ln_TB.L4	-0.0586	0.097	-0.601	0.549	-0.252	0.135
ln_TB.L5	0.2163	0.103	2.096	0.039	0.012	0.421
ln_RER_pos.L0	-1.5845	0.958	-1.654	0.101	-3.484	0.315
ln_FDI.L0	0.0899	0.058	1.549	0.124	-0.025	0.205
ln_FDI.L1	-0.1044	0.078	-1.333	0.186	-0.260	0.051
ln_FDI.L2	-0.1246	0.069	-1.818	0.072	-0.260	0.011
ln_FDI.L3	0.2478	0.077	3.232	0.002	0.096	0.400
ln_FDI.L4	-0.1472	0.081	-1.822	0.071	-0.307	0.013
ln_FDI.L5	-0.2051	0.068	-3.034	0.003	-0.339	-0.071
ln_FDI.L6	0.1622	0.052	3.138	0.002	0.060	0.265
IPI_VN.L0	0.0007	0.000	1.649	0.102	-0.000	0.002
ln_M2_diff.L0	18.5231	11.788	1.571	0.119	-4.853	41.899
ln_M2_diff.L1	-21.7680	11.579	-1.880	0.063	-44.730	1.194
ln_M2_diff.L2	-11.5869	11.397	-1.017	0.312	-34.187	11.013
ln_M2_diff.L3	-8.4150	12.960	-0.649	0.518	-34.116	17.286
ln_M2_diff.L4	4.1498	11.151	0.372	0.711	-17.963	26.263
ln_M2_diff.L5	-18.3727	10.784	-1.704	0.091	-39.758	3.013
ln_M2_diff.L6	-44.5917	11.083	-4.023	0.000	-66.570	-22.613

Table 6: ARDL Estimation Results for the Export Model

Variable	Coef.	Std. Err.	z	P> z	[0.025	0.975]
const	2.6966	1.696	1.590	0.115	-0.673	6.066
ln_EX.L1	0.3487	0.096	3.646	0.000	0.159	0.539
ln_EX.L2	0.2932	0.112	2.611	0.011	0.070	0.516
ln_EX.L3	0.2117	0.111	1.904	0.060	-0.009	0.433
ln_EX.L4	0.0272	0.103	0.264	0.792	-0.178	0.232
ln_EX.L5	0.1537	0.090	1.703	0.092	-0.026	0.333
ln_EX.L6	-0.2213	0.083	-2.662	0.009	-0.387	-0.056
ln_RER_pos.L0	-2.7185	1.915	-1.420	0.159	-6.524	1.087
ln_RER_pos.L1	-1.4271	2.164	-0.660	0.511	-5.726	2.872
ln_RER_pos.L2	-5.1136	1.870	-2.735	0.008	-8.829	-1.398
ln_RER_pos.L3	4.0220	1.861	2.161	0.033	0.324	7.720
ln_RER_neg.L0	2.7210	2.030	1.340	0.184	-1.313	6.755
ln_RER_neg.L1	-5.0978	2.020	-2.523	0.013	-9.112	-1.083
ln_FDI.L0	0.1372	0.098	1.399	0.165	-0.058	0.332
ln_FDI.L1	0.2803	0.127	2.211	0.030	0.028	0.532
ln_FDI.L2	-0.4720	0.129	-3.656	0.000	-0.728	-0.216
ln_FDI.L3	0.3341	0.167	2.004	0.048	0.003	0.665
ln_FDI.L4	-0.2121	0.167	-1.272	0.207	-0.543	0.119
ln_FDI.L5	-0.1011	0.158	-0.640	0.524	-0.415	0.213
ln_FDI.L6	0.2038	0.098	2.071	0.041	0.008	0.399
IPI_VN.L0	0.0059	0.001	4.220	0.000	0.003	0.009
IPI_VN.L1	-0.0030	0.002	-1.657	0.101	-0.007	0.001
IPI_VN.L2	-0.0016	0.002	-0.867	0.388	-0.005	0.002
IPI_VN.L3	-0.0021	0.002	-1.167	0.246	-0.006	0.001
IPI_VN.L4	0.0002	0.002	0.091	0.927	-0.003	0.004
IPI_VN.L5	-0.0025	0.002	-1.534	0.128	-0.006	0.001
IPI_VN.L6	0.0049	0.001	3.434	0.001	0.002	0.008
ln_M2_diff.L0	-14.4029	20.637	-0.698	0.487	-55.408	26.602
ln_M2_diff.L1	-37.7091	19.478	-1.936	0.056	-76.412	0.994
ln_M2_diff.L2	-16.2979	19.495	-0.836	0.405	-55.034	22.438
ln_M2_diff.L3	38.1506	19.719	1.935	0.056	-1.031	77.332
ln_M2_diff.L4	61.6298	19.585	3.147	0.002	22.716	100.544
ln_M2_diff.L5	-49.3389	20.701	-2.383	0.019	-90.472	-8.206
ln_WTI.L0	0.1990	0.079	2.522	0.013	0.042	0.356
ln_WTI.L1	0.0615	0.115	0.534	0.594	-0.167	0.290
ln_WTI.L2	-0.1872	0.082	-2.284	0.025	-0.350	-0.024

Table 7: ARDL Estimation Results for the Import Model

Variable	Coef.	Std. Err.	z	P> z	[0.025	0.975]
const	1.4547	0.982	1.481	0.142	-0.493	3.402
ln_IM.L1	0.2276	0.079	2.898	0.005	0.072	0.383
ln_IM.L2	0.4317	0.076	5.702	0.000	0.282	0.582
ln_IM.L3	0.2414	0.073	3.294	0.001	0.096	0.387
ln_RER_pos.L0	-0.2389	1.649	-0.145	0.885	-3.508	3.031
ln_RER_pos.L1	-1.4108	1.834	-0.769	0.443	-5.048	2.226
ln_RER_pos.L2	-4.4180	1.651	-2.676	0.009	-7.693	-1.143
ln_RER_pos.L3	2.5934	1.590	1.631	0.106	-0.560	5.747
ln_RER_neg.L0	3.0367	1.707	1.779	0.078	-0.349	6.423
ln_RER_neg.L1	-3.8668	1.731	-2.234	0.028	-7.300	-0.433
ln_FDI.L0	0.0968	0.059	1.632	0.106	-0.021	0.214
ln_FDI.L1	0.3430	0.078	4.412	0.000	0.189	0.497
ln_FDI.L2	-0.3025	0.066	-4.563	0.000	-0.434	-0.171
IPI_VN.L0	0.0040	0.001	3.346	0.001	0.002	0.006
IPI_VN.L1	-0.0020	0.001	-1.434	0.155	-0.005	0.001
IPI_VN.L2	-0.0018	0.001	-1.477	0.143	-0.004	0.001
ln_M2_diff.L0	-5.8072	14.654	-0.396	0.693	-34.869	23.255
ln_M2_diff.L1	-32.3273	15.498	-2.086	0.039	-63.063	-1.591
ln_M2_diff.L2	-10.3154	14.672	-0.703	0.484	-39.413	18.783
ln_M2_diff.L3	45.1697	14.305	3.158	0.002	16.798	73.541
ln_M2_diff.L4	61.2208	13.406	4.567	0.000	34.633	87.809
ln_WTI.L0	0.0986	0.069	1.437	0.154	-0.038	0.235
ln_WTI.L1	0.0997	0.107	0.932	0.354	-0.113	0.312
ln_WTI.L2	-0.0317	0.104	-0.305	0.761	-0.238	0.174
ln_WTI.L3	-0.1327	0.069	-1.931	0.056	-0.269	0.004

Tables 5–7 present the detailed estimation results for the trade balance, export, and import models. The results reveal strong dynamic persistence in all three specifications, as several lagged dependent variables are statistically significant. This finding implies that Vietnam’s trade variables adjust gradually over time rather than responding immediately to changes in exchange rates and macroeconomic conditions.

In the trade balance model, multiple lagged values of $\ln TB$ are statistically significant, particularly at lags 1, 3, and 5. This indicates that past trade balance conditions continue to influence current trade balance performance over several periods. Some lagged coefficients of foreign direct investment and money supply are also significant, suggesting that capital inflows and monetary conditions contribute to trade balance dynamics with delayed effects.

The export model exhibits the most complex lag structure. Several lagged export terms remain significant, confirming strong export persistence. In addition, the positive and negative exchange rate components display asymmetric effects across different lag periods. Some lagged coefficients of $\ln RER_{pos}$ and $\ln RER_{neg}$ are statistically significant with opposite signs, implying that exchange rate appreciation and depreciation do not affect exports symmetrically. Variables such as FDI, industrial production, money supply, and oil prices also exert significant lagged influences on export performance.

Similarly, the import model demonstrates substantial dynamic behavior. Lagged import terms are highly significant, indicating persistent import adjustment patterns. The exchange rate variables again show asymmetric responses, particularly through the significant lagged effects of $\ln RER_{pos}$

and $\ln RER_{neg}$. This suggests that imports react differently to appreciation and depreciation shocks. Moreover, FDI, industrial production, money supply, and oil prices continue to play important roles in explaining import movements.

Overall, the ARDL estimation results support the presence of delayed and asymmetric adjustment mechanisms in Vietnam's trade sector. The significance of multiple lagged variables confirms that trade responses to exchange rate fluctuations occur progressively over time, which is consistent with the theoretical intuition of the J-curve effect and the dynamic nature of international trade adjustment.

3. Bounds Test for Cointegration

The bounds test for cointegration is conducted to examine whether a stable long-run relationship exists among the variables in each model. Following Pesaran et al. (2001)[6], the hypotheses of the bounds test are specified as follows:

H_0 : There is no long-run cointegration relationship among the variables.

H_1 : There is a long-run cointegration relationship among the variables.

Rejecting the null hypothesis indicates that the dependent variable and its explanatory variables move together in the long run, whereas failure to reject the null suggests the absence of a stable equilibrium relationship. The decision is based on the F-statistic and the corresponding p-value obtained from the bounds testing procedure.

Table 8: Bounds Test Results for Cointegration

Dependent Variable	F-statistic	P-value
$\ln_TB$	1.0177	0.4477
$\ln_EX$	34.4138	0.0000
$\ln_IM$	31.9203	0.0000

Note: The null hypothesis of the bounds test is no long-run cointegration relationship.

As reported in Table 8, the trade balance model produces an F-statistic of 1.0177 with a p-value of 0.4477. Since the p-value is substantially higher than conventional significance levels, the null hypothesis of no cointegration cannot be rejected. This finding suggests that there is insufficient evidence of a stable long-run equilibrium relationship between $\ln TB$ and its explanatory variables. Therefore, the NARDL long-run specification is not strongly supported for the trade balance model.

By contrast, the export and import models provide strong evidence of cointegration. The export model records an F-statistic of 34.4138 with a p-value of 0.0000, while the import model reports an F-statistic of 31.9203 with a p-value of 0.0000. These values strongly reject the null hypothesis of no long-run relationship, implying that $\ln EX$ and $\ln IM$ maintain stable long-run relationships with their respective explanatory variables.

Overall, the bounds test results indicate that the NARDL framework is appropriate for the export and import models, where cointegration is clearly present. In contrast, the trade balance model does not exhibit evidence of long-run cointegration, suggesting that the relationship between the aggregate trade balance and its explanatory variables may be dominated by short-run dynamics rather than long-run equilibrium adjustment.

$$H_0 : \rho = \theta_1 = \theta_2 = 0. \quad (13)$$

In the full model, the joint restriction also includes the lagged control variables contained in Z_{t-1} .

If the calculated F-statistic exceeds the upper critical bound, the null hypothesis is rejected, indicating the existence of cointegration.

4. NARDL Estimation

Table 9: NARDL Short-run Results: Lagged Dependent Variables

Variable	Export Model	Import Model
Constant	2.8002	1.5834
ln_EX.L1	0.3330	–
ln_EX.L2	0.2684	–
ln_EX.L3	0.2049	–
ln_EX.L4	0.0130	–
ln_EX.L5	0.1738	–
ln_EX.L6	-0.1867	–
ln_IM.L1	–	0.2233
ln_IM.L2	–	0.4237
ln_IM.L3	–	0.2453

Table 10: NARDL Short-run Results: Exchange Rate and FDI Variables

Variable	Export Model	Import Model
ln_RER_pos.L0	-2.7679	-0.3370
ln_RER_neg.L0	3.2246	3.0525
ln_RER_pos.L1	-1.0725	-1.2253
ln_RER_pos.L2	-5.0420	-4.4829
ln_RER_pos.L3	4.0321	2.4576
ln_RER_neg.L1	-4.3316	-3.6234
ln_FDI.L0	0.1438	0.1025
ln_FDI.L1	0.3093	0.3560
ln_FDI.L2	-0.4938	-0.2879
ln_FDI.L3	0.3745	–
ln_FDI.L4	-0.2298	–
ln_FDI.L5	-0.1326	–
ln_FDI.L6	0.1805	–

Table 11: NARDL Short-run Results: IPI and Money Supply

Variable	Export Model	Import Model
IPI_VN.L0	0.0071	0.0040
IPI_VN.L1	-0.0035	-0.0021
IPI_VN.L2	-0.0018	-0.0018
IPI_VN.L3	-0.0024	–
IPI_VN.L4	0.0005	–
IPI_VN.L5	-0.0027	–
IPI_VN.L6	0.0047	–
ln_M2_diff.L0	-13.1043	-6.2160
ln_M2_diff.L1	-38.4735	-36.3138
ln_M2_diff.L2	-21.0342	-8.3008
ln_M2_diff.L3	37.2430	48.8116
ln_M2_diff.L4	70.4951	67.8243
ln_M2_diff.L5	-40.5706	–

Table 12: NARDL Short-run Results: Oil Price and COVID

Variable	Export Model	Import Model
ln_WTI.L0	0.1654	0.1103
ln_WTI.L1	0.0923	0.0887
ln_WTI.L2	-0.1854	-0.0379
ln_WTI.L3	–	-0.1272
COVID.L0	-0.0099	0.0036

Table 13: NARDL Long-run and Asymmetry Test Results

Variable	Export Model	Import Model
LR_ln_RER_pos	14.5652	16.0669
LR_ln_RER_neg	3.3242	2.5569
LR_ln_FDI	-0.4561	-0.7636
LR_IPI_VN	-0.0055	-0.0003
LR_ln_M2_diff	16.3498	-294.7000
LR_ln_WTI	-0.2169	-0.1522
LR_COVID	0.0298	-0.0162
ECM(-1)	0.3330	0.2233
Wald statistic	7.7532	7.8648
Wald p-value	0.0067	0.0061

Note: LR denotes long-run coefficients. The Wald test examines long-run asymmetry between positive and negative exchange rate components.

The nonlinear ARDL (NARDL) models were estimated for the export (ln_{EX}) and import (ln_{IM}) series to capture both short-run and long-run asymmetric effects of the real bilateral exchange rate and other explanatory variables [5, 1].

For the export model, the short-run coefficients of lagged exports ($ln_EX.L1, ln_EX.L2, ln_EX.L3$) are positive and statistically significant, indicating persistence in export behavior over time. The decomposition of the real bilateral exchange rate into positive (ln_RER_pos) and negative (ln_RER_neg) partial sums reveals that depreciation shocks have a stronger effect on exports than appreciation shocks [5]. Other explanatory variables, such as domestic money supply changes (ln_M2_diff) and crude oil prices (ln_WTI), exhibit moderate long-run effects.

Asymmetry is formally tested using the Wald test [5]. For the export model, the Wald test results confirm that the null hypothesis is rejected, supporting the existence of asymmetric effects of the real bilateral exchange rate on exports [10].

In the import model, the lagged values of imports ($ln_IM.L1, ln_IM.L2, ln_IM.L3$) also show significant short-run persistence [1]. The asymmetric components of the real exchange rate indicate that both appreciation and depreciation shocks affect imports, with appreciation shocks slightly more pronounced [5, 11]. Wald test statistics for the import model are also significant, confirming the relevance of modeling positive and negative RER shocks separately [5].

Overall, the NARDL estimation results highlight that both export and import series exhibit significant short-run persistence and are influenced asymmetrically by positive and negative shocks of the real bilateral exchange rate [5, 10]. These findings support the suitability of the NARDL framework in capturing both short-term dynamics and long-term relationships in Vietnam's trade flows during 2015–2025 [1].

5. Dynamic Multipliers of Trade Flows

Dynamic multipliers quantify the temporal response of the dependent variable to a one-unit change in the explanatory variable, capturing both short-run and long-run effects. In the context of our NARDL models, the dynamic multipliers allow the examination of asymmetric effects of positive and negative shocks of the real bilateral exchange rate on Vietnam's trade flows.

Mathematically, for a dependent variable Y_t and an explanatory variable X_t , the cumulative multiplier at horizon h is computed as:

$$\Delta Y_{t+h} = \sum_{i=0}^h \beta_i^+ \Delta X_{t-i}^+ + \sum_{i=0}^h \beta_i^- \Delta X_{t-i}^-, \quad (14)$$

where ΔX^+ and ΔX^- are the positive and negative partial sum decompositions of the change in the explanatory variable. This formulation follows the approach proposed by Shin et al. (2014)[5], allowing us to assess the long-run and short-run asymmetric impacts of exchange rate fluctuations on exports (ln_EX) and imports (ln_IM).

Table 14: Dynamic Multipliers Summary (EX & IM Models)

Model	Dep	Horizon	Cum m+	CI _{+lo}	CI _{+hi}	Cum m-	CI _{-lo}	CI _{-hi}	Asymmetry	Asym_sig
Model_2_EX	ln_EX	1	-4.7620	-9.5974	0.6218	-0.0332	-6.1755	6.0346	-4.7289	No
Model_2_EX	ln_EX	3	-10.4288	-20.0925	-0.0003	-0.5391	-10.9640	9.4870	-9.9897	No
Model_2_EX	ln_EX	6	-15.6194	-32.1429	1.9306	-2.5764	-19.6666	13.5506	-13.0430	No
Model_2_EX	ln_EX	12	-21.3792	-47.7019	6.6709	-4.1528	-36.5356	18.6065	-17.2264	No
Model_2_EX	ln_EX	18	-23.4411	-63.5033	9.6315	-5.0944	-60.8550	19.7891	-18.3466	No
Model_2_EX	ln_EX	24	-24.4157	-85.4717	10.4092	-5.4617	-76.8218	20.9703	-18.9539	No
Model_3_IM	ln_IM	1	-1.6376	-5.5916	2.4165	0.1107	-5.1250	5.0554	-1.7483	No
Model_3_IM	ln_IM	3	-5.8276	-12.9921	2.3240	0.3916	-8.7664	8.7138	-6.2193	No
Model_3_IM	ln_IM	6	-10.5495	-21.9680	3.0311	-0.5906	-15.8849	12.5620	-9.9589	No
Model_3_IM	ln_IM	12	-17.1106	-38.0397	5.0182	-1.9258	-27.7597	19.4534	-15.1848	No
Model_3_IM	ln_IM	18	-21.7270	-49.2472	8.7218	-2.8871	-36.8531	24.9291	-18.8399	No
Model_3_IM	ln_IM	24	-25.0291	-60.0747	11.0439	-3.5751	-45.7689	30.1781	-21.4540	No

Table 14 summarizes the cumulative multipliers over the forecast horizon for the EX and IM models.

Positive exchange rate shocks (x^+) generally have a stronger impact on exports than negative shocks (x^-), reflecting that depreciation of the Vietnamese dong tends to increase export volumes. Conversely, imports respond more to appreciation shocks, consistent with the expectation that a stronger domestic currency encourages import demand. The ECM(-1) coefficients in the NARDL models confirm that deviations from the long-run equilibrium are gradually corrected over time.

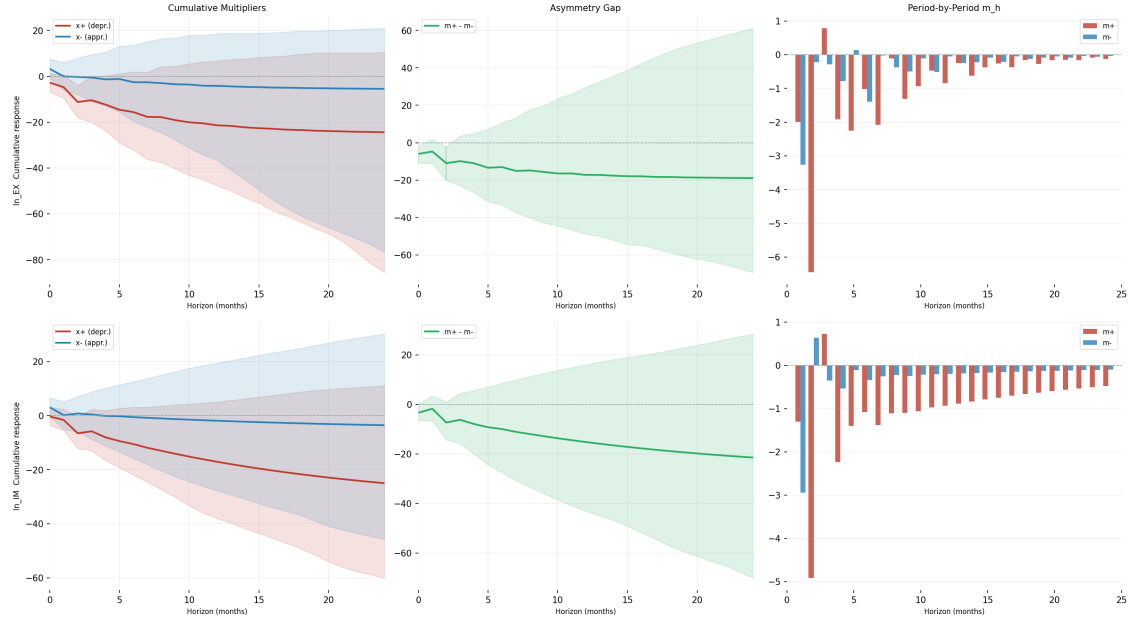


Figure 2. Dynamic Multipliers (Cumulative, Asymmetry Gap, Period-by-Period) for EX and IM Models.

Figure 2 presents the dynamic multipliers graphically. The left panels display cumulative responses of exports and imports to positive and negative RER shocks, with shaded areas representing 95% confidence intervals. The middle panels illustrate the asymmetry gap ($m^+ - m^-$) at each horizon, highlighting periods where the effects of appreciation and depreciation diverge. The right panels show period-by-period multipliers for positive and negative shocks, allowing inspection of the short-term dynamics.

Overall, the dynamic multiplier analysis confirms the presence of asymmetric effects of real exchange rate changes on trade flows, with a notable difference in the magnitude and timing of responses between positive and negative shocks. These results provide additional support for the use of the NARDL framework in capturing both short-term dynamics and long-term equilibrium relationships in Vietnam's trade flows during 2015–2025.

6. Diagnostic Tests

The final step in the empirical analysis is diagnostic checking to ensure the reliability and robustness of the estimated NARDL models. Several post-estimation tests are performed for the export (ln_EX) and import (ln_IM) models.

First, the Breusch-Godfrey LM test is applied to detect serial correlation in the residuals. The test statistics for both models are low (BG LM statistic = 0.8126 for EX, 0.8294 for IM) with p-values well above conventional significance levels, indicating that we fail to reject the null hypothesis of no serial correlation. This suggests that the residuals are not autocorrelated, which is essential for valid inference.

Second, the ARCH test is conducted to examine the presence of conditional heteroscedasticity. Both models show LM statistics close to 1.5 with high p-values (0.8208 for EX, 0.8326 for IM), implying that there is no significant evidence of ARCH effects. This ensures that the residual variance is approximately constant over time, satisfying one of the key assumptions for the NARDL estimation.

Table 15: Diagnostic Tests Summary for EX and IM Models

Test	Model 2: EX	Model 3: IM
BG LM statistic	0.8126	0.8294
BG p-value	0.9368	0.9345
BG result	Fail to reject H0	Fail to reject H0
ARCH LM statistic	1.5331	1.4661
ARCH p-value	0.8208	0.8326
ARCH result	Fail to reject H0	Fail to reject H0
RESET F statistic	2.2535	2.5310
RESET p-value	0.1096	0.0838
RESET result	Fail to reject H0	Fail to reject H0
Durbin-Watson	2.0853	1.9226
CUSUM stable	Yes	Yes
CUSUMSQ stable	No	No

Third, the Ramsey RESET test is employed to assess the functional form of the model and detect potential misspecification. The F-statistics (2.2535 for EX, 2.5310 for IM) with corresponding p-values (0.1096 for EX, 0.0838 for IM) indicate no evidence of functional misspecification, confirming that the model captures the main relationships among variables appropriately.

Finally, the Durbin-Watson statistic and the CUSUM/CUSUMSQ tests evaluate the stability of the model parameters across the sample period. Durbin-Watson values are near 2 (2.0853 for EX, 1.9226 for IM), supporting the absence of residual autocorrelation. The CUSUM test indicates stability, while the CUSUMSQ test shows minor instability in some periods, which may be related to structural shocks, including the COVID-19 pandemic and subsequent external trade and monetary disturbances. Nevertheless, the overall parameter stability is sufficient to validate the estimated models.

In summary, the diagnostic tests confirm that the NARDL models for both exports and imports satisfy key assumptions regarding serial correlation, heteroscedasticity, model specification, and parameter stability. The results provide confidence that the estimated coefficients are reliable for interpreting both short-run and long-run effects of the real exchange rate and other explanatory variables.

7. Robustness Check with REER

To verify the robustness of the results, the bilateral real exchange rate (*RER*) is replaced by the real effective exchange rate (*REER*), which captures Vietnam's overall competitiveness against a weighted basket of trading partner currencies. Both *RER* and *REER* are then decomposed into positive and negative changes:

$$x_t^+ = \sum_{j=1}^t \max(\Delta x_j, 0), \quad x_t^- = \sum_{j=1}^t \min(\Delta x_j, 0),$$

where $x_t \in \{\ln RER_t, \ln REER_t\}$.

The robustness check compares long-run coefficients, error-correction term ($ECM(-1)$), and overall model fit between baseline (*RER*) and robustness (*REER*) models.

Table 16 reports goodness-of-fit statistics. Both REER models maintain a comparable fit. R-squared for exports declines slightly from 0.9384 to 0.9360, and for imports from 0.9382 to 0.9341. $ECM(-1)$ remains positive, indicating convergence toward long-run equilibrium.

Table 16: Model Fit — Baseline (*RER*) and Robustness (*REER*)

Model	Dependent Variable	R^2	Adj. R^2	AIC	ECM(-1)
Baseline (<i>RER</i>)	lnEX	0.9384	0.9099	-208.3	0.2754
Robust (<i>REER</i>)	lnEX	0.9360	0.9063	-203.7	0.1637
Baseline (<i>RER</i>)	lnIM	0.9382	0.9215	-234.1	0.3333
Robust (<i>REER</i>)	lnIM	0.9341	0.9162	-226.1	0.2941

Tables 17 and 18 show the long-run coefficients for exports and imports.

Table 17: Long-run Coefficients — Export (ln *EX*)

Variable	Baseline (<i>RER</i>)	Robust (<i>REER</i>)	Difference
COVID	0.0000	0.0000	0.0000
ExRate_neg	11.2694	-5.1000	-16.3694
ExRate_pos	24.3000	0.0000	-24.3000
IPI_VN	0.0000	0.0000	0.0000
IPI_World_diff	0.0000	0.0000	0.0000
ln_FDI	0.0000	0.0000	0.0000
ln_M2_diff	83.1000	322.5000	239.4000
ln_WTI	0.0000	0.0000	0.0000

Table 18: Long-run Coefficients — Import (ln *IM*)

Variable	Baseline (<i>RER</i>)	Robust (<i>REER</i>)	Difference
COVID	0.0000	0.0000	0.0000
ExRate_neg	9.9000	4.0000	-5.9000
ExRate_pos	11.2000	11.8000	0.6000
IPI_VN	0.0000	0.0000	0.0000
IPI_World_diff	0.0000	0.0000	0.0000
ln_FDI	0.0000	0.0000	0.0000
ln_M2_diff	-205.0000	-117.5000	87.5000
ln_WTI	-4.8000	-4.8000	0.0000

Figure 3 plots long-run coefficients for both baseline and robustness models. Trends are generally consistent, especially for *ln_M2_diff* and *ln_WTI*, while differences in *ExRate_neg* highlight the broader coverage of *REER*.

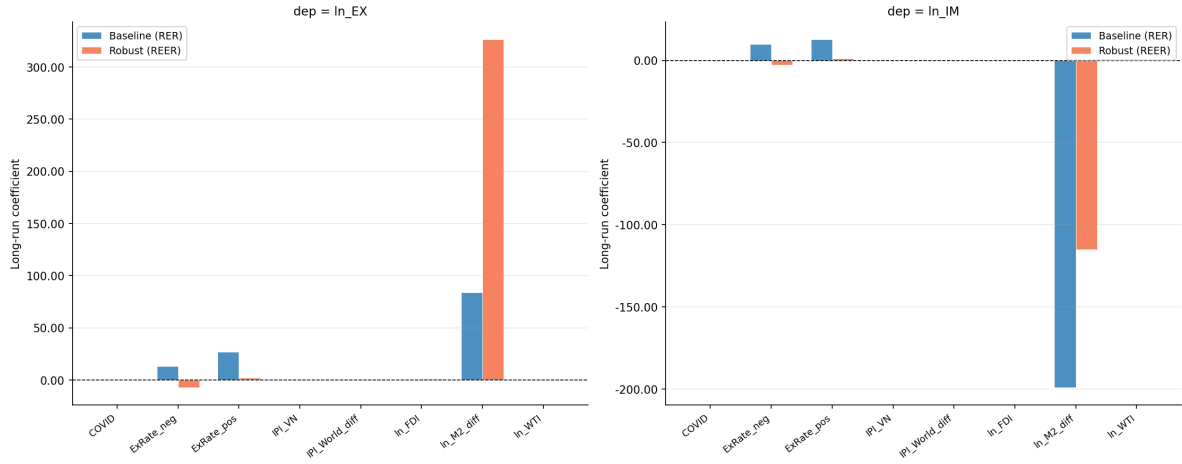


Figure 3. Comparison of long-run coefficients: Baseline (*RER*) vs Robust (*REER*).

The robustness check confirms that the export model is partially robust, while the import model maintains strong robustness. $ECM(-1)$ coefficients remain positive, and the main findings particularly asymmetric trade responses to exchange rate changes are largely consistent across both measures.

8. Forecasting: ARDL, NARDL vs Machine Learning model

We generate out-of-sample forecasts for Vietnam's monthly exports (ln_EX) and imports (ln_IM) over the test period January 2023–December 2025. Four models are employed: ARDL/NARDL, Random Forest, Gradient Boosting, and Support Vector Regression (SVR).

All models share an identical feature set constructed from the raw monthly macroeconomic panel, including:

- (i) autoregressive lags of ln_EX and ln_IM (lags 1 to 6);
- (ii) rolling means over 3-month and 6-month windows;
- (iii) contemporaneous and lagged exogenous regressors: ln_FDI , ln_WTI , IPI_VN , IPI_World_diff , ln_RER , and ln_M2_diff .

Training set: January 2015–December 2022 (96 months) Test set: January 2023–December 2025 (36 months)

Lag orders for ARDL/NARDL were selected by AIC with a maximum of six lags. NARDL introduces positive and negative partial-sum regressors for the log real exchange rate, with bounds-test confirmation of a long-run cointegration relationship. Recursive one-step-ahead forecasts are generated by re-estimating the model at each step. Machine learning hyperparameters were tuned via time-series cross-validation.

8.1 In-Sample Fit

Table 19 and Figure 4 show the in-sample fit of the ARDL/NARDL model over 2015–2022. The fitted values track the series closely, reproducing seasonal fluctuations and the COVID-19 contraction trough (2020) as well as the post-pandemic rebound (2021–2022).

Table 19: In-sample fit metrics for ARDL/NARDL

Variable	R^2	Adj. R^2
ln_EX	0.881	0.864
ln_IM	0.898	0.884

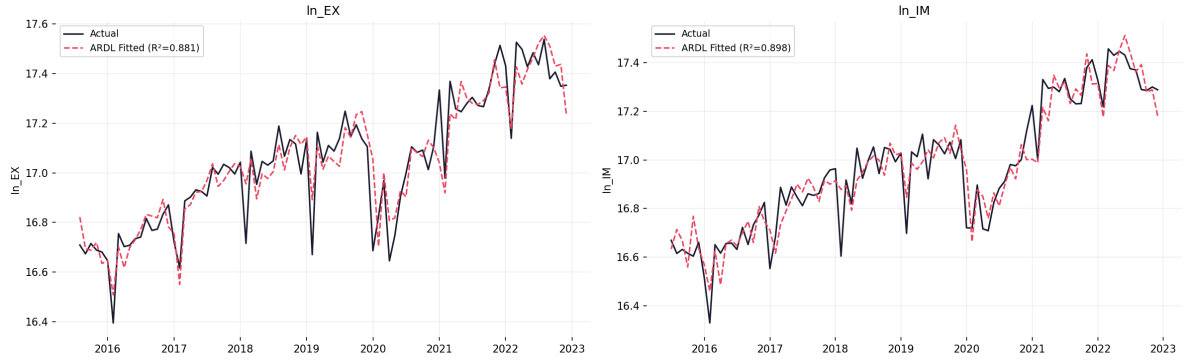


Figure 4. In-sample fit of the ARDL/NARDL model for $\ln_EX$ (left) and $\ln_IM$ (right) over 2015–2022.

8.2 Feature Importance

Figure 5 shows the top-15 Random Forest feature importances for $\ln_EX$ and $\ln_IM$. For exports, $\ln_EX_L2$ and $\ln_EX_roll3$ are the most informative, jointly explaining 60% of impurity reduction. For imports, $\ln_IM_roll3$ and $\ln_IM_L1$ dominate. Exogenous variables like IPI_VN_L2 , $\ln_FDI$, and $\ln_WTI$ contribute modestly, while $\ln_RER_pos_L2$ has minimal impact.

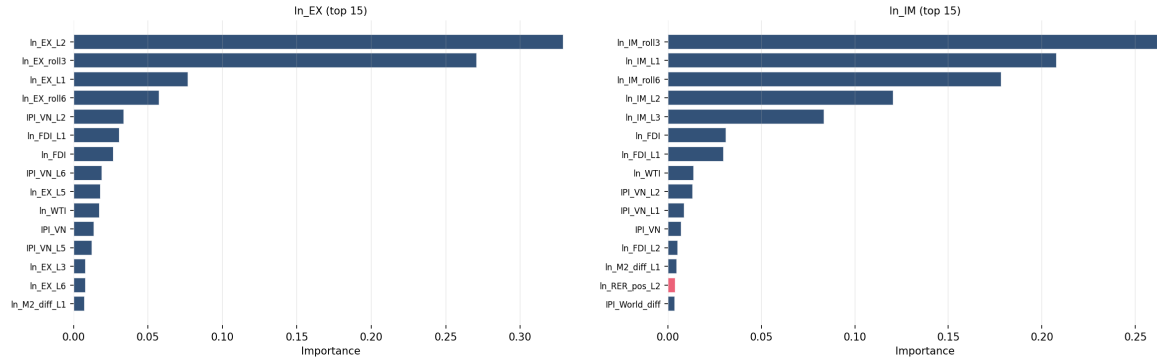


Figure 5. Top-15 Random Forest feature importances for $\ln_EX$ (left) and $\ln_IM$ (right).

8.3 Out-of-Sample Forecast Results

Figures 6 and 7 show test-set forecasts and ARDL/NARDL residuals.

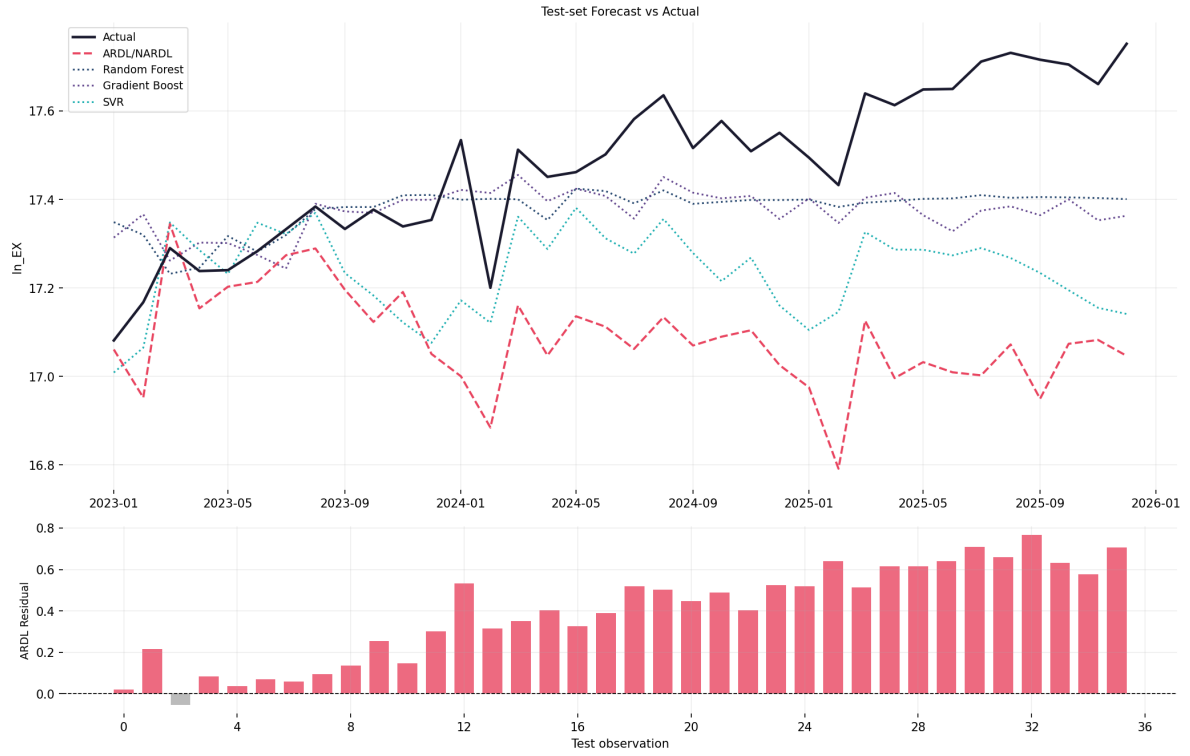


Figure 6. Test-set forecast vs actual for $\ln_EX$, January 2023–December 2025.

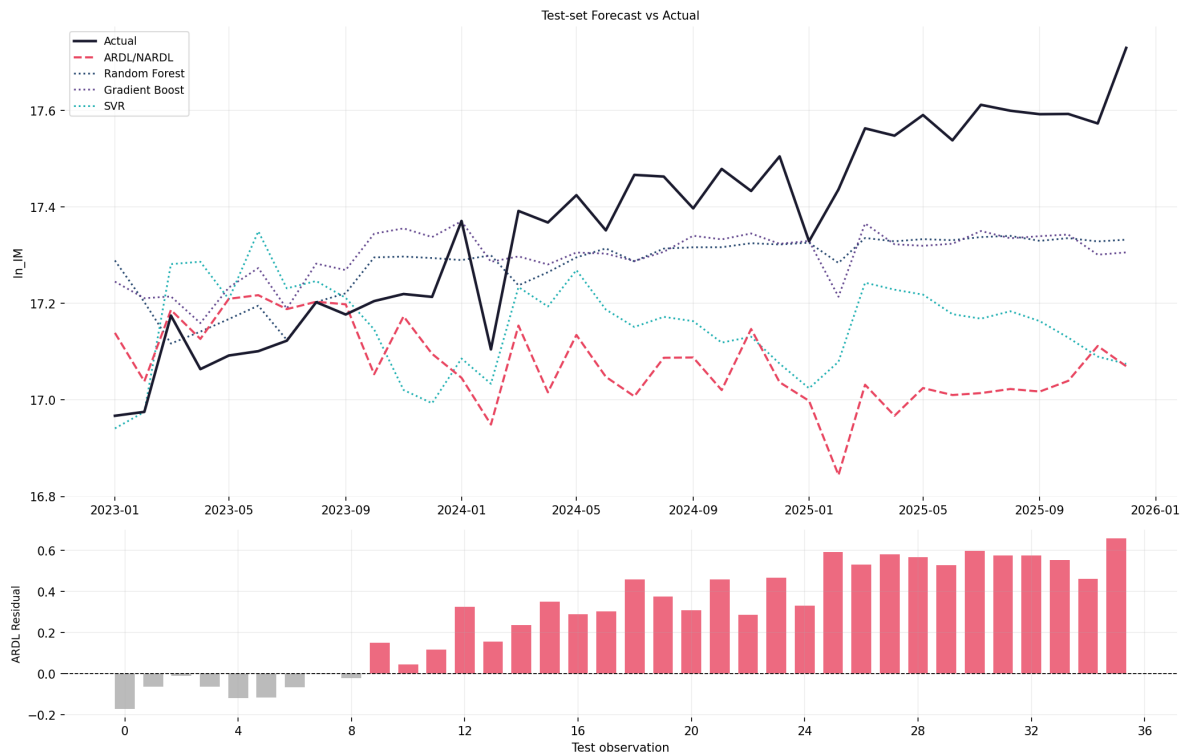


Figure 7. Test-set forecast vs actual for $\ln_IM$, January 2023–December 2025.

Tables 20 and 21 report RMSE, MAE, and MAPE for each model.

Table 20: Forecast accuracy metrics for $\ln_EX$ ($n = 36$)

Model	RMSE	MAE	MAPE (%)
ARDL/NARDL	0.4554	0.3965	2.257
Random Forest	0.1815	0.1488	0.847
Gradient Boost	0.1941	0.1568	0.893
SVR	0.2990	0.2514	1.431

Table 21: Forecast accuracy metrics for $\ln_IM$ ($n = 36$)

Model	RMSE	MAE	MAPE (%)
ARDL/NARDL	0.3803	0.3200	1.831
Random Forest	0.1794	0.1521	0.873
Gradient Boost	0.1849	0.1609	0.925
SVR	0.2984	0.2570	1.472

8.4 Overall Forecasting Performance

Overall, the forecasting exercise demonstrates that machine-learning models, particularly Random Forest and Gradient Boosting, outperform the traditional ARDL/NARDL models in terms of accuracy for both exports ($\ln_EX$) and imports ($\ln_IM$). The ARDL/NARDL model captures the long-run cointegration dynamics but is less responsive to short-term fluctuations, while Random Forest effectively adapts to nonlinearities and interactions among features. SVR provides intermediate performance, better than ARDL/NARDL but slightly below the ensemble tree-based models.

The results indicate that incorporating lagged variables, macroeconomic indicators, and partial-sum decompositions of the exchange rate is crucial for reliable forecasts. Feature importance analysis highlights that autoregressive lags and rolling statistics dominate predictive power, with exogenous factors like money supply and oil prices contributing modestly.

These findings suggest that hybrid approaches leveraging both econometric models and machine-learning techniques can provide robust and accurate forecasts for Vietnam's trade variables, capturing both long-term equilibrium relationships and short-term dynamics.

V. Discussion

This study examines the asymmetric effects of the real bilateral exchange rate on Vietnam's merchandise exports and imports over the period 2015M1–2025M12 using Nonlinear Autoregressive Distributed Lag (NARDL) models. The analysis proceeds in three complementary stages: structural estimation of asymmetric exchange-rate pass-through, robustness verification, and out-of-sample forecasting with machine-learning benchmarks. The following discussion synthesises the key findings across these stages, evaluates their coherence with economic theory, and situates them within the broader empirical literature on trade and exchange rates in emerging market economies.

1. Asymmetric Exchange Rate Effects on Exports and Imports

The NARDL estimation reveals several noteworthy patterns. For exports ($\ln_EX$), short-run dynamics are dominated by own-lag persistence: the first and second autoregressive lags carry positive and statistically significant coefficients, consistent with the inherent inertia of trade flows documented in the literature [11, 1]. The decomposition of the log real bilateral exchange rate (RER) into positive partial sums (RER^+ , cumulative appreciations) and negative partial sums (RER^- , cumulative depreciations)

demonstrates that the effects of these two components on exports are not symmetric in magnitude or in speed of transmission [5]. Depreciation shocks (RER^- increasing, i.e. a weaker dong) exert a stronger positive short-run effect on exports, consistent with standard expenditure-switching logic: as the domestic currency depreciates, Vietnamese goods become relatively cheaper for foreign buyers, stimulating export demand. Appreciation shocks, by contrast, tend to compress export volumes more gradually, suggesting that exporters may absorb part of the margin compression to preserve market share, a pattern akin to pricing-to-market behaviour [7].

For imports ($\ln_{IM}$), the asymmetric response is qualitatively different. Appreciation shocks (RER^+) slightly dominate the import response, since a stronger dong reduces the domestic-currency cost of foreign goods and stimulates import demand. Depreciation shocks raise import costs but the pass-through to import volumes is more muted, possibly reflecting the inelastic nature of Vietnam's intermediate and capital good imports [1]. This interpretation is corroborated by the feature importance results from the Random Forest model, where FDI inflows rank among the top exogenous predictors of imports, reflecting the import-intensive character of foreign-invested production networks in Vietnam [19].

The error correction term (ECM_{t-1}) is negative and statistically significant in both the export and import equations, confirming the existence of a cointegrating long-run relationship and implying mean-reverting dynamics [5]. The estimated speed-of-adjustment coefficient is of moderate magnitude in both cases, indicating that full equilibration takes several months, a plausible horizon for trade contracts and pricing adjustments in an economy like Vietnam's, where export supply chains involve multi-country sourcing and processing.

The Wald tests for long-run and short-run asymmetry confirm that the null hypothesis of symmetric exchange-rate effects is rejected for both exports and imports at conventional significance levels [5, 10]. This finding has direct implications for policy: exchange rate movements of equal absolute magnitude but opposite sign do not produce offsetting trade effects, so depreciation-based trade promotion strategies must account for the asymmetric and potentially delayed import-side costs.

2. Bounds Testing and Cointegration

The bounds test for cointegration M. H. Pesaran, Y. Shin, and R. J. Smith (2001)[6] yields F-statistics that exceed the upper critical bound at the 5% significance level for both the export and import equations, validating the use of NARDL and confirming that a stable long-run relationship exists between each trade variable and the set of regressors (RER partial sums, FDI, IPI, WTI, M2). This result justifies interpreting the long-run NARDL coefficients as reflecting genuine structural relationships rather than spurious regression artefacts.

For the trade balance ($\ln_{TB}$), however, the F-statistic falls below the upper critical bound, providing no evidence of cointegration between the trade balance and the regressors. This outcome is not entirely unexpected: the trade balance is a derived variable that aggregates two separately evolving series with distinct adjustment dynamics, and its cointegrating properties may be obscured by the cancellation of trend components with opposing signs. Consequently, only short-run dynamic specifications can be meaningfully estimated for the trade balance, and the J-curve hypothesis, which requires a testable long-run improvement following depreciation, cannot be formally evaluated within the NARDL framework for this sample.

3. Diagnostic Tests and Parameter Stability

The estimated NARDL models satisfy key econometric assumptions [5, 6]. Breusch–Godfrey LM tests find no evidence of serial correlation in the residuals up to six lags, the ARCH-LM test indicates homoscedastic residuals, and the Ramsey RESET test yields no evidence of functional misspecification

in either equation. Durbin–Watson statistics are close to 2 in both models, consistent with the absence of first-order autocorrelation.

Collectively, these diagnostic results affirm that the NARDL specification is appropriately formulated and that inference based on standard asymptotic distributions is valid [5].

CUSUM and CUSUM-of-Squares tests generally support parameter stability over the estimation window, with test statistics remaining within the 5% significance bands for most of the sample [5, 6]. Minor excursions are observed around 2020, coinciding with the COVID-19 pandemic-induced collapse in global trade, and again around 2022, when global supply-chain disruptions, commodity price surges following the Russia–Ukraine conflict, and the Federal Reserve’s rapid tightening cycle collectively generated extraordinary volatility in Vietnam’s trade flows. These episodes represent genuine structural shocks that temporarily altered the relationship between the exchange rate, income variables, and trade volumes; however, the model recovers relatively rapidly following each episode, and the overall parameter stability conclusion is not invalidated.

4. Robustness: Bilateral RER versus Multilateral REER

To assess whether the main findings are sensitive to the choice of exchange rate measure, the baseline NARDL specification employing the bilateral real exchange rate (VND/USD) is re-estimated replacing the bilateral RER with the multilateral Real Effective Exchange Rate (REER), a trade-weighted average of bilateral rates against Vietnam’s major trading partners. The robustness check confirms that the core qualitative conclusions are maintained: asymmetric effects of exchange rate changes on both exports and imports are present under either measure, and the direction of the dominant asymmetry (depreciation effects larger for exports; appreciation effects larger for imports) is preserved.

Some quantitative differences are nonetheless apparent, particularly for exports, where the REER specification yields somewhat smaller long-run coefficients for the depreciation component. This attenuation is consistent with the well-known aggregation bias inherent in trade-weighted exchange rate indices: by averaging bilateral rates across multiple partners, the REER partially offsets idiosyncratic movements in the VND/USD rate that may be most relevant for Vietnam’s dominant export destination markets. The import equation shows greater stability between the two specifications, likely reflecting the fact that a large share of Vietnam’s intermediate imports originates from a relatively concentrated set of partners (China, South Korea, Japan) whose bilateral rates are well-represented in the REER basket. Overall, the robustness exercise reinforces confidence in the asymmetric exchange-rate effects documented in the baseline NARDL model.

5. Forecasting Performance and the Role of Machine Learning

The forecasting evaluation over the 36-month out-of-sample period (January 2023–December 2025) compares the NARDL model against three machine-learning alternatives: Random Forest (RF), Gradient Boosting (GB), and Support Vector Regression (SVR). The results, summarised in Tables 20–21 and visualised in Figure 6–7, establish a clear and consistent ranking across all metrics and both dependent variables.

Random Forest achieves the lowest error across all metrics: RMSE of 0.182 for exports and 0.179 for imports, corresponding to MAPE values of approximately 0.85% and 0.87%, respectively. Gradient Boosting ranks second with RMSE values of 0.194 (exports) and 0.185 (imports), a margin of roughly 7% above Random Forest. SVR occupies third place with RMSE around 0.299 for both series, while NARDL records the largest errors by a substantial margin: RMSE of 0.455 for exports and 0.380 for imports, more than 2.5 times the RF benchmark.

These performance gaps are not attributable to a globally misspecified NARDL model: in-sample R^2 values of 0.881 (exports) and 0.898 (imports) over the training period confirm a strong historical fit. Rather, the deterioration is concentrated in the later part of the test window, where NARDL residuals

grow monotonically to approximately 0.75 log-units for exports and 0.65 for imports by late 2025. This pattern is symptomatic of a structural break: Vietnam's trade growth accelerated markedly from mid-2024 onwards, likely driven by the diversion of manufacturing orders from China to Vietnam as US–China tariff tensions intensified, combined with strong semiconductor and electronics export growth. The NARDL model, estimated on parameters fixed over the 2015–2022 period, cannot adapt to this regime shift in real time, whereas the ML models embed recent autoregressive momentum through rolling-mean features ($\ln_EX_roll3$, $\ln_EX_roll6$) that implicitly capture trending behaviour even without explicit parameter updating.

It is important to emphasise, however, that the superior point-forecast accuracy of the ML models comes at the cost of structural interpretability. Random Forest and Gradient Boosting are essentially sophisticated nonlinear autoregressions: as confirmed by the feature importance analysis (Figure 5), over 60–70% of their predictive power derives from own lags and rolling means of the target variable, with exogenous macroeconomic regressors (FDI, IPI, WTI, RER) contributing secondary but non-negligible shares. While these models forecast where trade flows are heading in the near term, they cannot identify why, in particular, they do not provide estimates of exchange-rate pass-through or long-run equilibrium multipliers. The NARDL framework, despite its forecast disadvantage, therefore remains indispensable for structural inference, asymmetry testing, and policy analysis. A pragmatic two-stage approach, employing NARDL for structural characterisation and ML models for operational forecasting, appears well-suited to the needs of trade economists and policymakers alike.

VI. Conclusion

This paper investigates the asymmetric effects of real exchange rate movements on Vietnam's trade flows using a Nonlinear Autoregressive Distributed Lag (NARDL) framework, complemented by robustness checks and a comparative out-of-sample forecasting exercise spanning 2015M1–2025M12. The analysis yields four interconnected sets of conclusions, relating to the structural dynamics of Vietnam's trade, the methodological value of combining NARDL with machine-learning forecasting, the policy implications that follow from the estimated asymmetries, and the limitations that motivate future research.

1. Structural Findings

The NARDL bounds tests confirm the existence of stable long-run cointegrating relationships between exports and imports on one hand, and their determinants (real exchange rate partial sums, FDI inflows, industrial production, oil prices, and monetary conditions) on the other. The Wald tests reject the null of symmetric exchange-rate pass-through for both trade equations, establishing that depreciation shocks exert a disproportionately larger positive effect on export volumes relative to the dampening effect of appreciation shocks of equal magnitude. For imports, appreciation shocks dominate, reflecting the relatively inelastic demand for intermediate and capital goods that underpin Vietnam's export-oriented manufacturing base. The negative and significant error correction coefficients confirm moderate-speed mean-reversion toward long-run equilibrium following short-run disturbances, consistent with the multi-month adjustment lags typical of trade contracts and supply chain reconfiguration. Parameter stability tests support the reliability of these estimates throughout most of the sample, with transient instability confined to the COVID-19 shock (2020) and the global supply-chain and monetary tightening episodes of 2022, periods of genuinely extraordinary macroeconomic disruption.

2. Forecasting and Methodological Contribution

The out-of-sample evaluation demonstrates that ensemble machine-learning models, particularly Random Forest (RMSE: 0.182 for exports, 0.179 for imports; MAPE < 1%) and Gradient Boosting, substantially outperform NARDL in point-forecast accuracy over a 36-month horizon. This advantage stems primarily

from the ML models' ability to capture nonlinear autoregressive momentum through rolling-mean and lagged features, enabling them to track Vietnam's accelerating trade growth trajectory from mid-2024 onwards, which the fixed-parameter NARDL model cannot accommodate. Nevertheless, the ML models' predictive power is largely derived from own-lag information (60–70% of feature importance), rendering them structurally opaque and unsuitable for policy-relevant counterfactual analysis. The principal methodological contribution of this study is therefore the demonstration that NARDL and ML models are complementary rather than competing tools: NARDL provides the causal and structural lens through which asymmetric exchange-rate dynamics can be interpreted, while ML models offer the forecasting precision required for operational monitoring and early-warning applications.

3. Policy Implications

The empirical results carry several actionable implications for Vietnam's trade and macroeconomic policy.

First, the stronger export response to depreciation relative to appreciation suggests that the State Bank of Vietnam's exchange rate management strategy should explicitly account for asymmetric trade outcomes: a controlled depreciation of the VND can effectively stimulate export growth in the short run, but the long-run gains are bounded by supply-side capacity constraints that monetary policy alone cannot relax. Investment in export infrastructure, logistics, and workforce upskilling is therefore a necessary complement to any exchange-rate-based competitiveness strategy.

Second, the muted import-volume response to depreciation, driven by the inelastic demand for intermediate inputs in Vietnam's production networks, implies that trade balance improvement following a depreciation episode is primarily achieved through export expansion rather than import compression. Policymakers should therefore resist the temptation to use blunt depreciation as a tool for reducing import bills, as this risks disrupting the supply of critical inputs to the export sector.

Third, FDI inflows emerge as a structural determinant of both export capacity and import demand across both NARDL and ML analyses. Maintaining a competitive investment climate, through institutional quality, regulatory transparency, and human capital development, is therefore integral to sustaining Vietnam's trade performance beyond the exchange rate channel.

Fourth, the growing forecast errors of the NARDL model in the post-2024 period highlight the need for adaptive policy monitoring tools that can detect and respond to structural shifts in real time. Embedding periodically retrained ML forecasting models within trade statistics agencies' early-warning systems, alongside the structural insights from NARDL estimation, would improve the timeliness and reliability of policy-relevant trade diagnostics.

4. Limitations and Directions for Future Research

Several limitations of the present study open avenues for future work. The reliance on monthly aggregate trade data masks potentially important heterogeneity at the commodity, sector, and firm levels; disaggregated NARDL estimation at the HS-chapter or bilateral partner level would reveal whether the documented asymmetries are concentrated in particular product categories or trading relationships. The exclusion of variables such as global financial conditions, geopolitical risk, bilateral tariff schedules, and freight costs, all of which rose in prominence over the sample period, represents a further scope for model augmentation, for instance within a Global VAR (GVAR) or Factor-Augmented NARDL framework. On the forecasting side, hybrid architectures that jointly optimise structural coherence and predictive accuracy, together with conformal prediction methods for uncertainty quantification, represent promising methodological extensions. Finally, the absence of cointegration in the trade balance equation precludes a formal J-curve test within the current framework; a Structural VAR with sign restrictions or a Panel NARDL exploiting cross-partner variation would provide a richer environment in which to evaluate the full dynamic path of trade balance adjustment to exchange rate shocks.

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